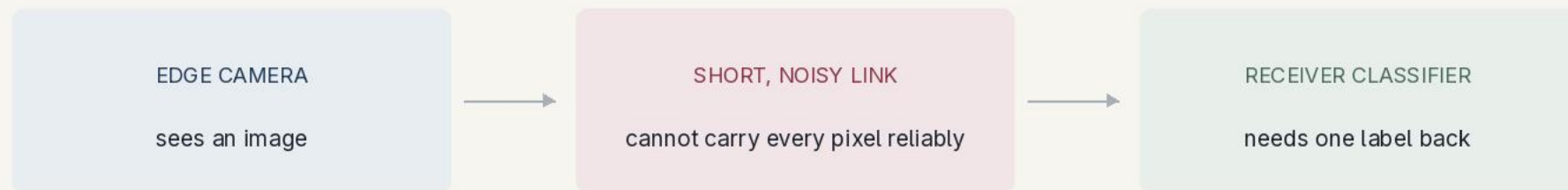


Task-Oriented Deep Joint Source–Channel Coding

What a camera should send when the receiver only needs the answer



THESIS

Train the sender and the receiver together so the classification decision survives a short, noisy link.

Second Review, 29 September – 3 October 2026.
Everything here was measured on held-back pictures. The exam set has not been opened.

What has changed since the first review

In August we had a working digital system and no trained learned model. We have both now.

Aug	The digital system	Measured how badly it loses pictures at every setting, tested it on all 288,000 held-back combinations, and froze the settings it should use at each signal strength
Aug	The learned system	Built the training loop, trained models at several seeds, and froze three of them at half rate
Sep	The fair comparison	Built the control that sends learned features digitally, so we are not just comparing against a normal compressed picture
Sep	Weak-link training	Trained a second version with the signal strength picked at random, and compared it against the first
Sep	The crossing question	Decided it on held-back pictures before seeing the answer: the two lines cross between -5 and -4 dB
Sep	Power limits	Trained one model under a hard limit on how tall the signal spikes may get
Oct	The measurements	252 measurements across twelve versions of the system and twenty-one signal strengths, 1,000 held-back pictures each
Nov	The exam set	Still locked. It opens in week 11 once the locking document is signed

Everything above ran on the validation split. The test split stays sealed until gate G-12.

Objectives, stated as completion criteria

The rubric asks whether we finished what we set out to do, so we wrote it as things we did, not results we hoped for.

WHAT WE SET OUT TO DO

- 1 Build it** Train a learned sender and a learned receiver together, so the image goes through the same kind of noise a radio would add.
- 2 Give both the same radio time** At every signal strength and every rate, the two systems get the same number of chances to use the radio.
- 3 Charge the overhead** Checksums, padding and error-correction bytes all come out of the budget before any picture data does.
- 4 Pick the setting fairly** The operating point is chosen using the digital system alone, before anyone looks at the learned curve.
- 5 Report per picture** Every row records what happened to one picture, including the ones that never arrived.

WHAT WE DID NOT AIM FOR

“Show that the learned system beats the digital one.”

That is a result, not an objective. Our specification treats both a crossing and the learned system winning everywhere as complete, successful outcomes. The rubric scores the second review on what we set out to do, not on which way the curve went.

We found a crossing, so the fallback we had planned for never happened and the objectives were not rewritten.

Before the numbers, one picture

Every chart that follows runs from a very weak radio signal on the left to a clean one on the right.

PLAIN WORDS / OUR TERM

How strong the signal is	SNR, in decibels	the left-to-right axis
How many pictures got through	delivery coverage	the flat 10% line
How many we labelled correctly	top-1 accuracy	the up-and-down axis
How many times we use the radio	channel-use budget	12,800 or 3,200 per picture

Every point on every chart is the same 1,000 pictures, sent at that one signal strength.

ONE EXAMPLE

At the weakest signal we measured:

The learned system labelled 728 of the 1,000 pictures correctly.

The digital system got almost nothing through, so every picture fell back to one fixed answer — and that answer happens to be right 100 times out of 1,000. So 10%.

That flat 10% is not a bad prediction. It is 1,000 pictures that never arrived.

These were 1,000 held-back pictures. The exam pictures stay sealed until week 11.

What we measured for this review

We built twelve versions of the system and tested each one at twenty-one signal strengths.

252

individual measurements,
one row each in the
published record

12

versions of the system, five
of them controls

21

signal strengths, from very
weak to very clean

1,000

held-back pictures behind
every single point

0

exam pictures ever opened

RADIO TIME

Half rate — 12,800 uses of the radio per picture
Quarter rate — 3,200 uses per picture

Within one rate, both systems get the same amount of radio time. It does not mean they send the same amount of information, and the two sides carry different kinds of picture.

WHO MARKS IT

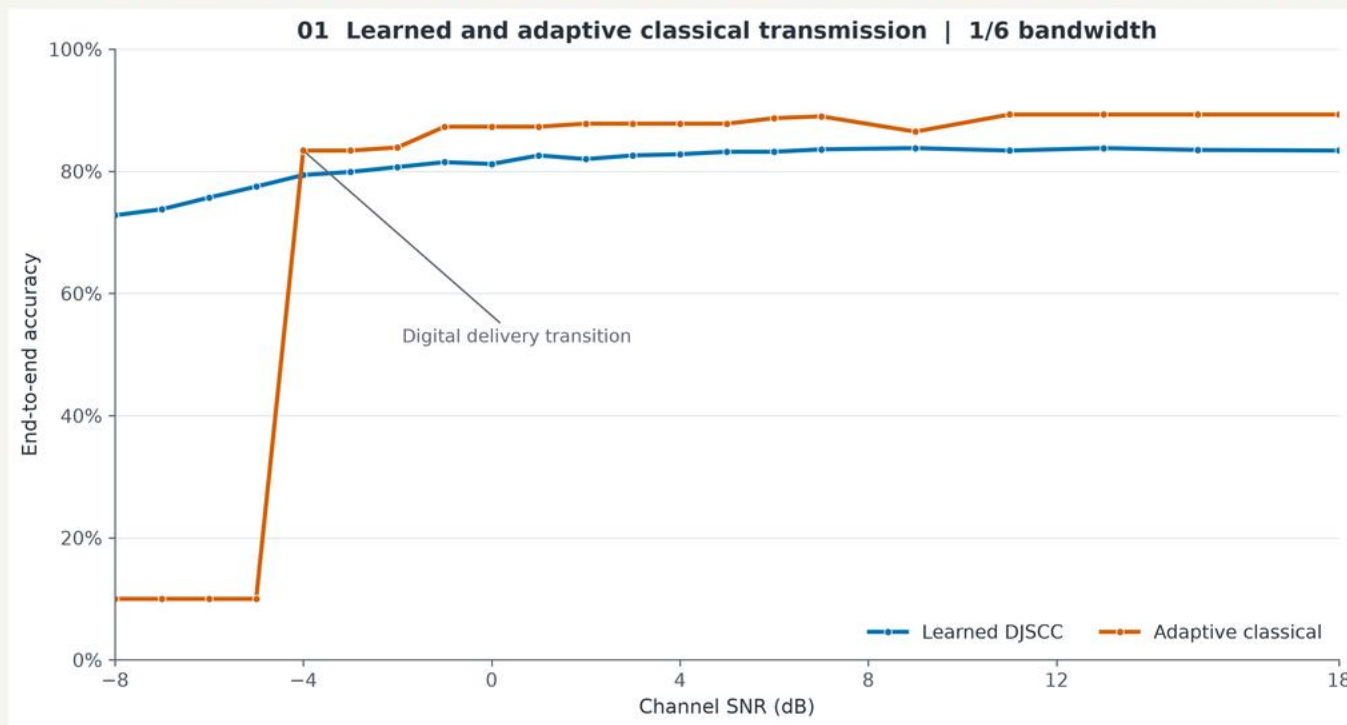
The learned systems are marked by the classifier built into them. The classical systems are marked by a classifier we trained specifically on compressed-looking images.

So when we compare the two, we are comparing whole systems — the link, the way the picture is represented, and the marker. We say so wherever the comparison appears.

One training run. These are observed differences, not confidence intervals.

Where the digital system gives up, and where it takes over

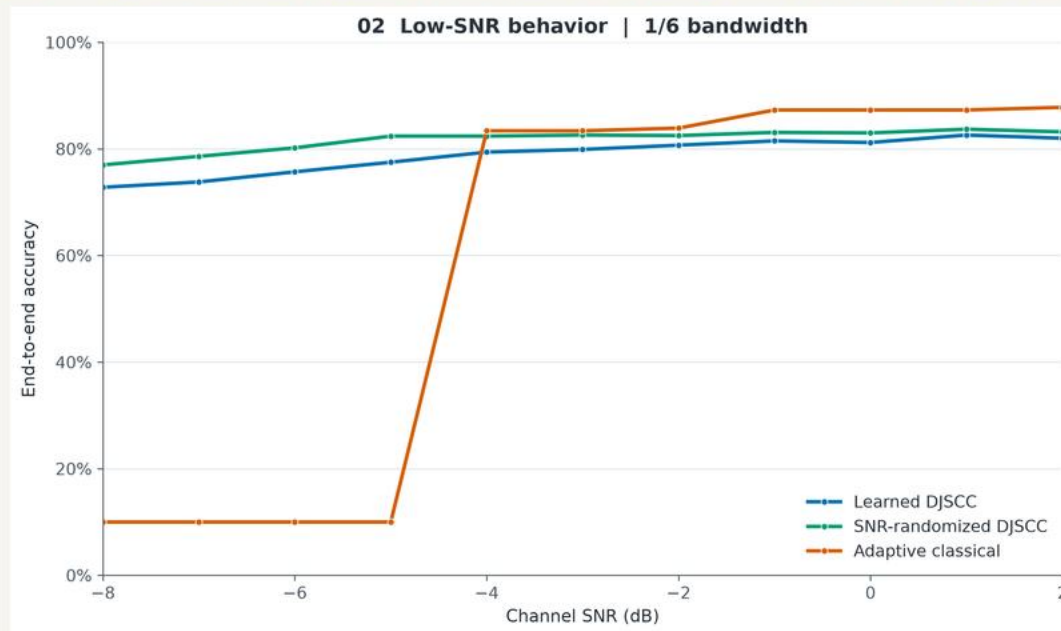
The learned system keeps answering the whole way across. The digital system stays silent, then overtakes it.



Weakest signal: learned right on 728 of 1,000, digital got nothing through. At -4 dB the digital system starts working and beats the learned one. At +18 dB it is well ahead.

Close-up: the moment the digital system starts working

Between -5 and -4 dB the digital system begins getting pictures through, and its score jumps 73 points.



At -8 dB

The learned system is right on 728 of 1,000. The digital system got nothing through, so it scores 10%.

At -4 dB

The digital system now gets everything through and reaches 83.4%, ahead of the learned system's 79.4%.

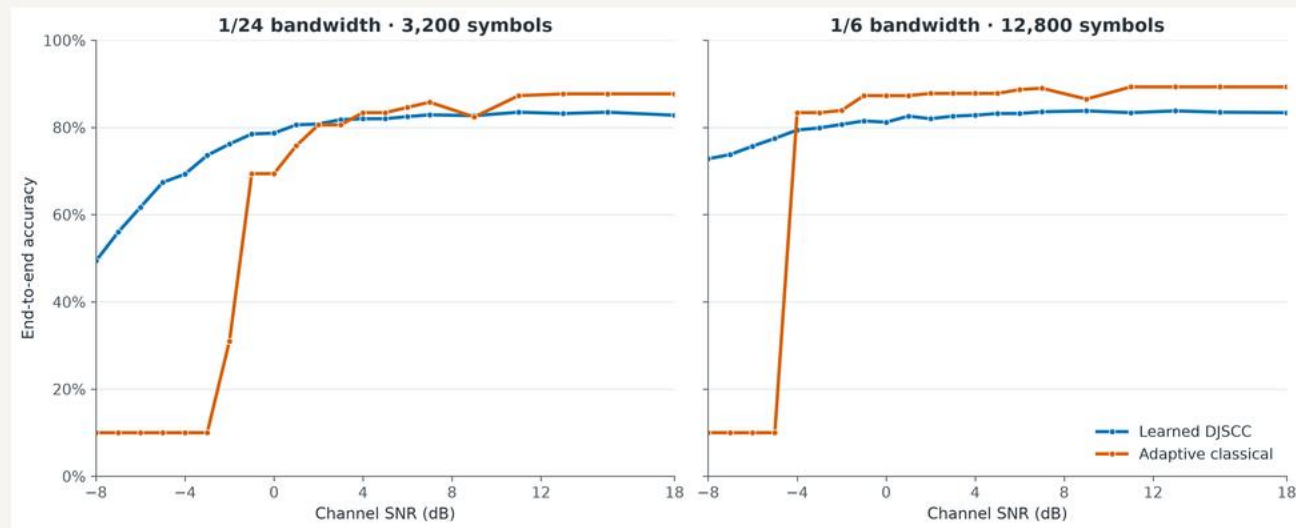
After that

It dips at -2 dB and again at $+9$ dB, so its line is not perfectly smooth.

We never measured a signal strength between -5 and -4 dB, so this is a gap between two points, not a known threshold.

What happens when we quarter the radio time

On a quarter of the radio time, the digital system loses far more at the weak end than the learned one does.



Weak signal, half → quarter

The learned system drops from 72.8% to 49.4% rightness. A 23 point loss.

At 0 dB, quarter rate

Learned 78.7%, digital 69.4%. The learned system is still ahead.

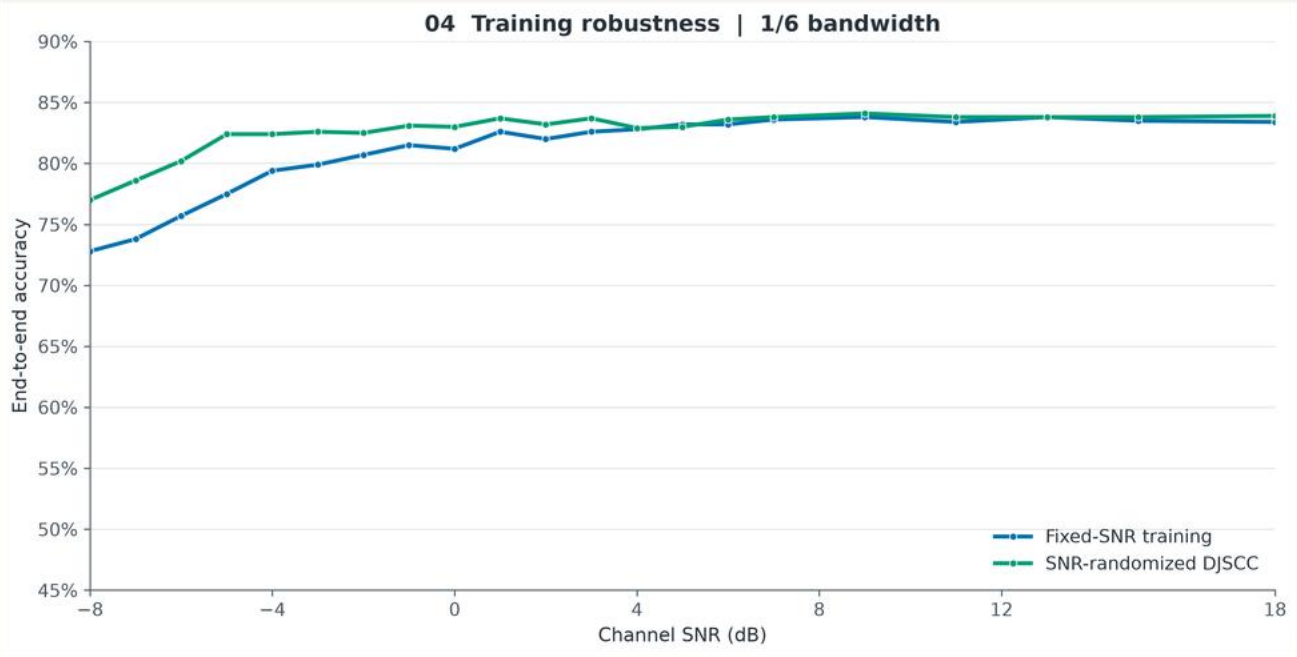
Digital catches up, then slips

Digital first passes the learned system at the +4 dB point, then loses the lead again at +9 dB where it drops some pictures.

Because it crosses and then uncrosses, we should not claim one clean crossover at this rate.

Training across signal strengths instead of one

Picking the training signal strength at random each time lifts the whole weak end of the line.



SYSTEM	-8 dB	-4 dB	0 dB	+9 dB	+18 dB
Random each time	77.0	82.4	83.0	84.1	83.9

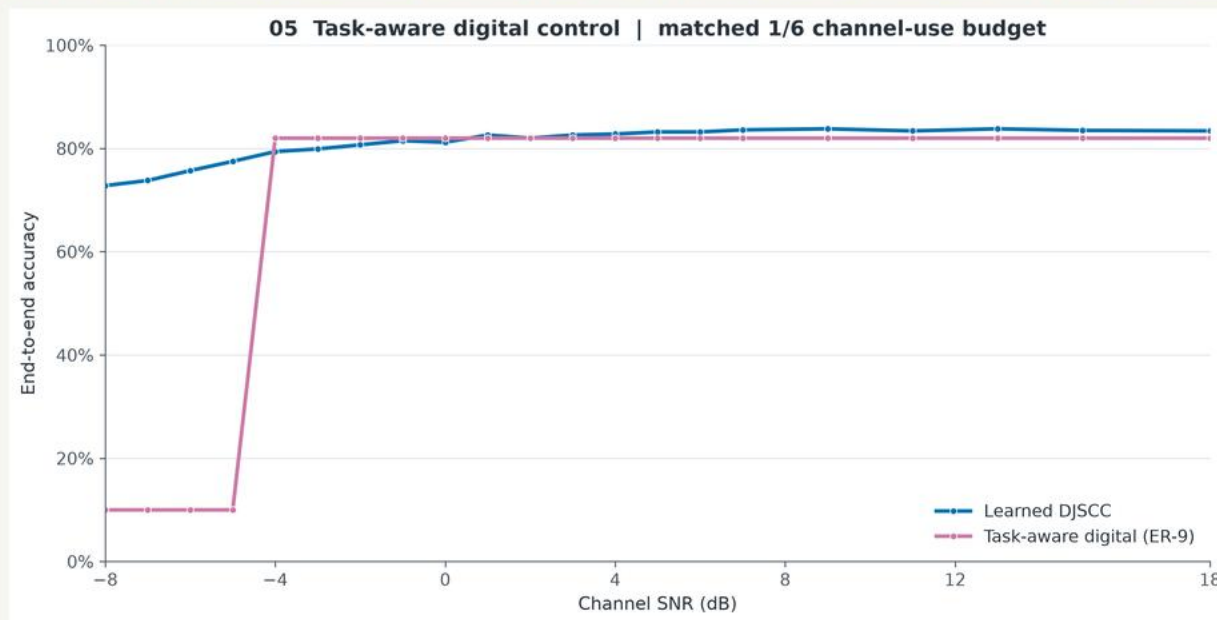
One fixed strength

Two separately trained models from one seed. This is not a measure of how much results vary between runs.

77.0 79.4 81.2 83.8 83.4

The control that keeps the comparison honest

If we had only compared against a normal compressed image, the weak-signal story would be too easy to believe.



What this arm does

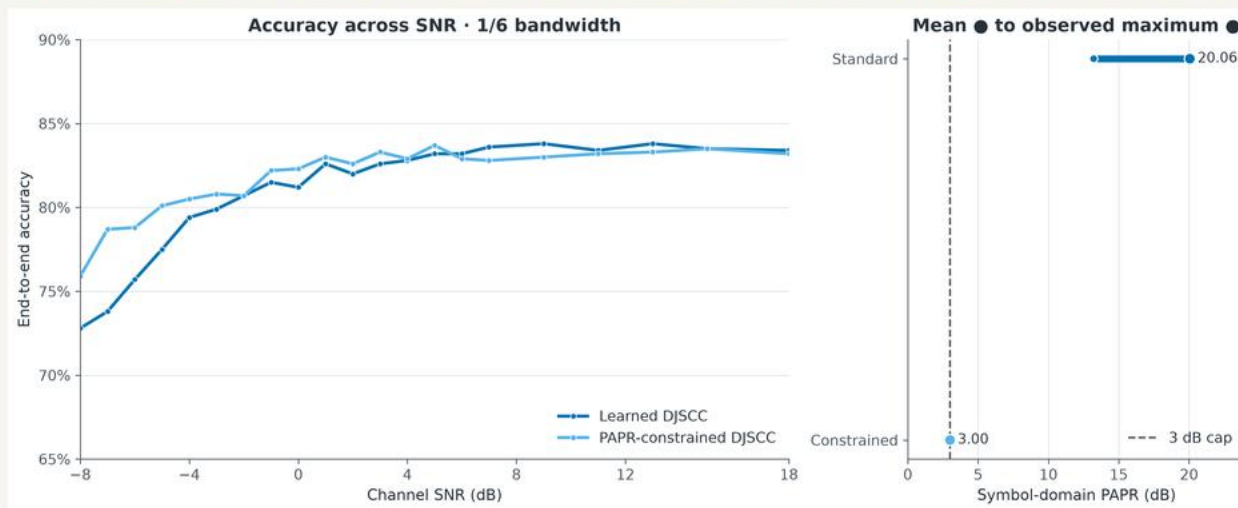
Instead of a JPEG image it sends the learned features themselves, through the same error-correction code and the same radio settings, using the same 12,800 radio uses. It is the control that separates "knows the task" from "codes both at once".

What it settles

It holds 82.0% at every strength from -4 dB up. Sending learned features digitally is strong once pictures get through, so the weak-signal gap is about failed deliveries as much as representation. It does not isolate the error-correction code.

Keeping the signal peaks from getting too tall

A real transmitter cannot send a signal with tall spikes, so we trained a second model under that limit.



“Peak-to-average power ratio” = how much taller the tallest spike is than the average

3.000002 dB

tallest spike, limited model — inside the cap

20.059284 dB

tallest spike, the ordinary learned model

+3.1 pp

better at -8 dB, limited versus ordinary

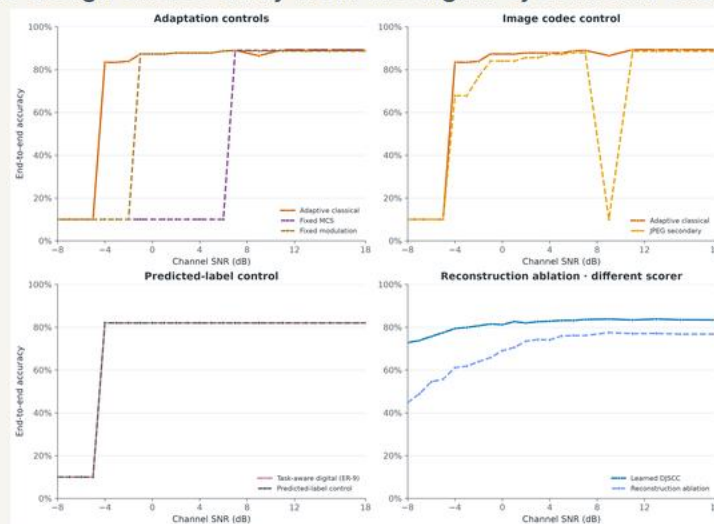
-0.2 pp

worse at +18 dB — the limit costs nothing up here

This measures the numbers we put on the radio, not a real amplifier. And the limited model is a separate training run, so the accuracy gap is not a clean penalty for the limit.

Four controls, including one that went against us

Each panel takes one design choice away from the digital system and shows what it was worth.



Fixed settings

No delivery until +6 dB, then 89.0%. Choosing new settings per signal strength buys real reach.

JPEG instead of JPEG 2000

88.6% at +18 dB, and a real point at +9 dB where nothing got through. We kept it.

Fixed modulation

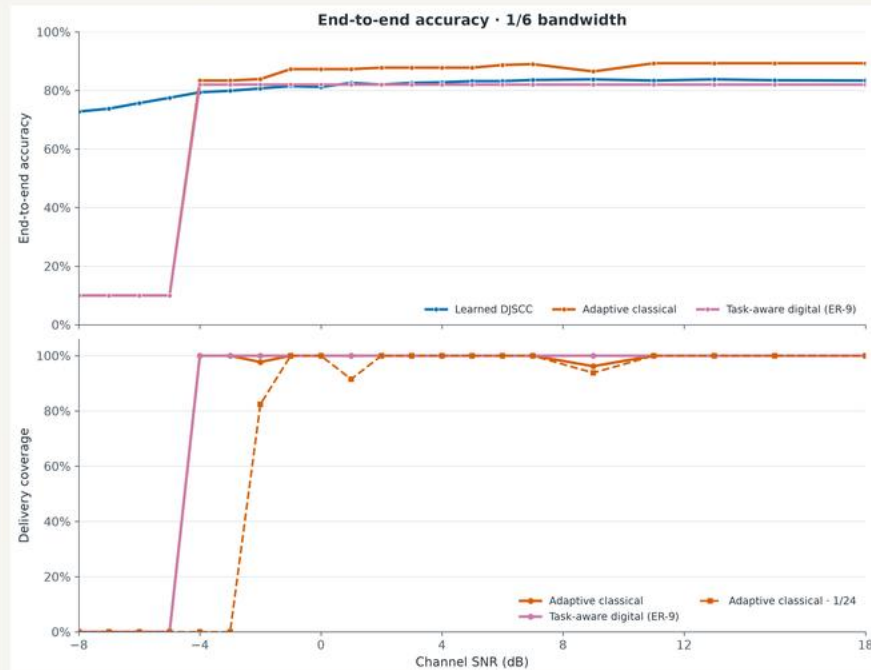
Starts working later than choosing per strength, and settles near 88.7%.

Sending the answer itself

82.0% once it arrives. A label the sender guessed, not the true answer.

Why the digital line breaks where it does

The score only moves when the pictures start or stop arriving.



READING IT

The flat 10% is a rule, not a delivery

When the digital system gets nothing through, every picture takes the same fallback answer. On an evenly split set that is right 100 times in 1,000.

Failed deliveries are part of the result

We report how many pictures arrived next to how many we got right, so you can tell a broken link from a wrong guess.

The learned system always answers

Its rules promise a label at every signal strength, which is why its line has no floor to sit on.

We did not add any points between the ones we measured, and we did not smooth any curve to hide a step.

The numbers behind the curves

How many of the 1,000 held-back pictures we labelled correctly, counted one picture at a time.

HALF RATE · 12,800 RADIO USES PER PICTURE

SYSTEM	-8	-5	-4	0	+9	+18
Learned, trained once	72.8	77.5	79.4	81.2	83.8	83.4
Digital, picks settings	10.0	10.0	83.4	87.3	86.5	89.3
Learned, random signal	77.0	82.4	82.4	83.0	84.1	83.9
Digital, learned features	10.0	10.0	82.0	82.0	82.0	82.0
Learned, power-limited	75.9	80.1	80.5	82.3	83.0	83.2

QUARTER RATE · 3,200 RADIO USES PER PICTURE

SYSTEM	-8	-4	-2	0	+4	+18
Learned, trained once	49.4	69.3	76.2	78.7	82.0	82.8
Digital, picks settings	10.0	10.0	31.0	69.4	83.4	87.7

Where the two lines cross

At half rate, which system is ahead flips between -5 dB and -4 dB. We decided the crossover question on exactly that, before seeing the answer.

WHAT THIS TABLE IS NOT

Twenty-one signal strengths is not twenty-one independent experiments. Every cell comes from the same 1,000 held-back pictures and the same single training seed, so a smooth-looking line tells you nothing statistically. A one or two point gap between rows is an observed difference and nothing more. The full per-picture records that would let us compute proper uncertainty ranges are not in this checkout, so we have not computed any.

What this evidence cannot tell you

One training run, held-back pictures only

Everything here is a description. No ranges, no significance, no repeated trials.

A different grader on each side

The learned systems are marked by their own built-in classifier, the digital systems by one we trained on compressed images. So this measures whole systems, not error-correction coding on its own.

Not the same amount of effort

The digital system's settings and its grader were frozen from earlier work. The learned versions are separate training runs. That is a specific, fair comparison, but not a matched-compute one.

No real amplifier involved

We measured how tall the spikes are in the numbers we send. Nothing here shows what a real transmitter would do. The project stays on a simulated channel.

Frozen choices make sharp steps

The fallback rule, 1,000 pictures and fixed operating points produce jumps and a few bumpy points. We did not smooth them.

Our specification treats both a crossing and the learned system winning everywhere as complete, successful outcomes. Neither result undoes the other.

Four times we changed the approach because of what we found

The rubric asks what we decided based on the results obtained. These are the four that count.

AM-34

We had pinned the digital system to the weakest radio setting, which made the two lines crossing impossible by arithmetic. We let it pick a stronger setting as the signal improves.

The rule we adopted: every change strengthens the digital system or is decided in advance. We never weaken the learned system.

AM-52

Our signal strengths were spaced 2 dB apart exactly where the sharpest drop happens, so we could have missed it entirely. We added three more strengths before it mattered.

Put the detail where the effect is, not where it is convenient to sample.

AM-58

The script that checks our packet sizes reported no problems while breaking four of its own rules. We fixed the script, not the results.

All 215 configurations stayed feasible and every scientific outcome came out the same.

AM-60

The exam pictures were set to unlock in week 9, about three weeks before the manifest that locks them existed. We moved that to week 11.

They have stayed locked ever since. Nothing has been read.

Twenty-two further changes are written down in SPEC §17, including three checks that turned out to be wrong and had to be repaired.

What we claim as new, and what we do not

Both claims are about how the experiment is built, not about the idea of sending meaning instead of pixels.

The task-aware digital control

A digital arm sends learned features through the same error-correction code and radio settings, using the same amount of radio time. It tells “knows the task” apart from “codes both at once” inside one experiment.

Different from: studies that report a learned-versus-digital gap without such a control.

Charging the overhead to the digital system

The picture file, framing, checksums and rate-matching bytes are all measured and counted, so the digital system is tuned on the payload it can genuinely use.

Different from: pipelines that quote a nominal rate and leave the overhead uncounted.

NOT CLAIMED AS NEW

- Getting steadily worse as the link worsens
- Avoiding the sudden drop-off
- Sending task-aware features digitally
- Training a neural encoder and decoder

People have already done all four of these. Claiming them would cost more than it earns.

NEGATIVE CHECK

Both claims above are checked against the 30 references in docs/literature-review.md, and any claim that something has never been reported has to trace back to that list rather than to our own specification. The full written statement ships with the final review package.

Where the remaining weeks went

W10		29 Sep – 3 Oct	Second Review	Held-back measurements complete; package and figures frozen	done
W11		5 – 11 Oct	Open the exam set	One guarded opening once the locking manifest is signed	planned
W12		12 – 18 Oct	Decide everything	Every hypothesis settled on the exam set; Tier 1 frozen	planned
W13		19 – 25 Oct	Demo	Signal-strength slider, both systems side by side, frozen plot	planned
W14		26 Oct – 1 Nov	Poster + optional hardware	Poster draft; real radio only if Tier 1 leaves room	planned
W15		2 – 8 Nov	Internal freeze	Report finished in the required format	planned
W16		9 – 15 Nov	Spare week	Report completion and checking only. No new experiments	planned
W17		16 – 22 Nov	Final Review	Final review 17–21 Nov; report and supporting material due 20 Nov	planned

The report is due inside Final Review week, which is why W15 is an internal freeze and W16 holds no experimental work.

Where this leaves the project

The learned system keeps answering when the link is too weak for the digital one to get a single picture through. Once the digital system starts delivering, it is the more accurate of the two. They cross between -5 and -4 dB.

Next

One guarded run on the exam set once the locking manifest is signed. So far, zero exam pictures opened.

Then

Every hypothesis settled on the exam set, and the first tier frozen.

Later

Real-radio replay and the live demo are stretch goals. The project does not depend on them.

EVIDENCE TRAIL

252-measurement record and per-picture manifest in `results/learned/w10/` · figures and data in `presentation-results/` · requirements and changes in `spec/SPEC.md §17`

The exam set has never been opened. It stays that way through every gate, run and record in this project.